STAT A411/611 Lecture Notes

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Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Lecture Plan and Review

Today: Chapter 2, Sections 2.1–2.3.

Wednesday: Sections 3.1 and 3.2.

Last time: Section 1.3, counting rules and probability applications.

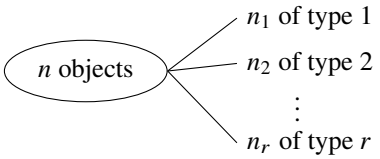
Review of Chapter 1

Section 1.1 introduced the **Classical Formula (C.F.)**; Section 1.2 introduced **set theory**. Section 1.3 used the **Basic Rule of Counting (BRC)** for permutations, combinations, and probability applications.

$$\text{Permutations of distinct objects:} \quad {}_nP_k = \frac{n!}{(n-k)!},$$

$$\text{Combinations:} \quad {}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

For permutations of indistinguishable objects, group the n objects by type:



$$\frac{n!}{n_1!n_2!\cdots n_r!}$$

2 Chapter 2 - Probability Concepts

2.1 Section 2.1 - Probability Axioms

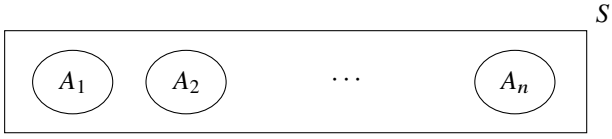
(A1) $P(A) \geq 0$ for any event A.

(A2) $P(S) = 1$.

(A3) If $A_1, A_2, \ldots, A_n$ are **mutually exclusive**, then

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = P(A_1) + P(A_2) + \cdots + P(A_n).$$

Mutually exclusive events cannot occur at the same time.

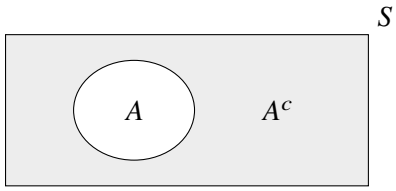


Rule 1: Complement Rule (CR)

For any event A:

The probability that A occurs is one minus the probability that A does not occur:

$$\underbrace{P(A)}_{\substack{\text{directly}\\ \text{LONG}}} = 1 - \underbrace{P(A^c)}_{\text{QUICKER}}.$$



Proof. Since $A \cap A^c = \varnothing$ and $A \cup A^c = S$, the events A and A^c are mutually exclusive. Thus

$$1 \overset{\text{A2}}{=} P(S) = P(A \cup A^c) \overset{\text{A3}}{=} P(A) + P(A^c),$$

which gives $P(A) = 1 - P(A^c)$. □

Example: At Most Three Aces

A dealer shuffles a deck of 52 cards and selects five cards without replacement for a player. What is the probability of receiving **at most three aces**? The cards are given to the **first player**. The five cards are selected **randomly**.

Direct approach. The four possible ace counts are mutually exclusive:

LONG M.E. $\longrightarrow$ A3.

$$\begin{aligned} P(\text{at most 3 aces}) &= P(0 \text{ aces OR } 1 \text{ ace OR } 2 \text{ aces OR } 3 \text{ aces}) \\ &\overset{\text{A3}}{=} \underbrace{P(0 \text{ aces}) + P(1 \text{ ace}) + P(2 \text{ aces}) + P(3 \text{ aces})}_{\text{four probabilities}}. \end{aligned}$$

or use C.R.

Complement approach. It is shorter to calculate the probability of all four aces:

$$\begin{aligned} P(\text{at most 3 aces}) &\overset{\text{CR}}{=} 1 - P(\text{more than 3 aces}) \\ &= 1 - \underbrace{P(\text{all 4 aces})}_{\text{one probability}} \\ &\overset{\text{C.F.}}{=} 1 - \frac{\binom{4}{4}\binom{48}{1}}{\binom{52}{5}} = 1 - \frac{1 \cdot 48}{52!/(5!\,47!)} \\ &= 1 - \frac{48}{\frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}} \\ &= 1 - \frac{\cancel{48} \cdot 120}{52 \cdot 51 \cdot 50 \cdot 49 \cdot \cancel{48}} = \boxed{1 - \frac{120}{52 \cdot 51 \cdot 50 \cdot 49}}. \end{aligned}$$

Quick!

Rule 2: General Addition Rule (GAR)

Let $A_1, A_2, \ldots, A_n$ be n events.

The **General Addition Rule**, also called the **inclusion-exclusion formula**, applies to events $A_1, A_2, \ldots, A_n$:

$$\begin{aligned} P(A_1 \cup A_2 \cup \cdots \cup A_n) = {} & P(A_1) + P(A_2) + \cdots + P(A_n) \\ & - P(A_1 \cap A_2) - P(A_1 \cap A_3) - \cdots - P(A_{n-1} \cap A_n) \\ & + P(A_1 \cap A_2 \cap A_3) + \cdots + P(A_{n-2} \cap A_{n-1} \cap A_n) \\ & - P(A_1 \cap A_2 \cap A_3 \cap A_4) - \cdots \\ & + \cdots + (-1)^{n+1} P(A_1 \cap A_2 \cap \cdots \cap A_n). \end{aligned}$$

Add the single-event probabilities, subtract pairwise intersections, add triple intersections, and continue with alternating signs. Each distinct intersection is counted once.

Two events ($n = 2$).

$$\boxed{P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).}$$

Three events ($n = 3$).

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) = {} & P(A_1) + P(A_2) + P(A_3) \\ & - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) \\ & + P(A_1 \cap A_2 \cap A_3). \end{aligned}$$

Example: A Heart or a Face Card

Shuffle a 52-card deck and select one card at random. What is the probability that the card is a heart or a face card? Define

$$H = \{\text{the selected card is a heart}\},$$

$$F = \{\text{the selected card is a face card}\}.$$

The face cards are the jack, queen, and king. In particular,

$$H \cap F = \{J_H, Q_H, K_H\}.$$

Therefore,

$$P(H \cup F) \overset{\text{GAR}}{=} P(H) + P(F) - P(H \cap F)$$

$$\overset{\text{C.F.}}{=} \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \boxed{\frac{22}{52}}.$$

The cards in the union can also be listed by suit:

$$\text{Hearts:} \quad A_H, 2_H, 3_H, \ldots, 10_H, \underbrace{J_H, Q_H, K_H}_{\text{also face cards}}$$

$$\text{Other face cards:} \quad J_S, Q_S, K_S; \quad J_D, Q_D, K_D; \quad J_C, Q_C, K_C.$$

The three heart face cards belong to both sets, so their probability is subtracted once.

Rule 3: Special Addition Rule (SAR)

Rule 3 = Axiom 3.

For **mutually exclusive** events, every intersection of two or more distinct events is empty. The General Addition Rule therefore reduces to Axiom 3:

$$\begin{aligned} P(A_1 \cup \cdots \cup A_n) &= P(A_1) + \cdots + P(A_n) \\ &\quad - 0 - 0 - \cdots + 0 + 0 + \cdots + (-1)^{n+1} 0 \\ &= \boxed{P(A_1) + P(A_2) + \cdots + P(A_n)}. \end{aligned}$$

2.2 Section 2.2 - Conditional Probabilities

The probabilities considered so far, such as $P(A)$, $P(A^c)$, $P(A \cup B)$, and $P(A \cap B)$, are **unconditional probabilities**. Let B be an event. Then $P(B)$ denotes the (unconditional) probability that B occurs.

Suppose new information tells us that event A has occurred. We update the probability of B:

$$P(B) \xrightarrow{\text{given that } A \text{ has occurred}} P(B \mid A)$$

Definition. $P(B \mid A)$ is the probability that B occurs *given that A has occurred.*

$$P(B \mid A): \quad \text{the probability of } B \text{ given } A.$$

Order matters. Union and intersection are commutative, but conditional probabilities are not generally equal when their events are reversed:

$$\begin{aligned} &P(A \cup B) = P(B \cup A), \qquad P(A \cap B) = P(B \cap A), \\ &P(B \mid A) \neq P(A \mid B) \quad \text{in general.} \end{aligned}$$

Example 1: A Fair Six-Sided Die

Without additional information, the possible outcomes are $1, 2, 3, 4, 5, 6$. Given an odd result, the remaining possibilities are $1, 3, 5$. Thus

$$P(\text{a five comes up}) = P(\boxed{\cdot\cdot}) \overset{\text{C.F.}}{=} \frac{1}{6},$$

$$P(\text{a five comes up} \mid \text{an odd \# comes up}) \overset{\text{C.F.}}{=} \frac{1}{3}, \qquad \{1, 3, 5\}.$$

Example 2: Two Cards Without Replacement

Shuffle a deck of 52 cards and randomly select two cards without replacement. The two conditional probabilities are

$$P(\text{second card is an ace} \mid \text{first card is not an ace}) \overset{\text{C.F.}}{=} \frac{4}{51},$$

$$P(\text{second card is an ace} \mid \text{first card is an ace}) \overset{\text{C.F.}}{=} \frac{3}{51}.$$

Rule 4: Conditional Probability Rule (CPR)

$$\boxed{P(B \mid A) = \frac{P(A \cap B)}{P(A)}} \qquad \big(P(A) > 0\big).$$

Example 2.2.1

Define the events

$$A = \{\text{the pregnancy zone protein is present in a pregnant woman}\},$$
$$M = \{\text{a randomly selected baby is male}\}.$$

The given probabilities are

$$P(A) = 0.43, \qquad P(M) = 0.51, \qquad P(M \cap A) = 0.17.$$

Find: $P(M \mid A)$.

Then

$$P(M \mid A) \overset{\text{CPR}}{=} \frac{P(M \cap A)}{P(A)} = \frac{0.17}{0.43} \approx \boxed{0.395}.$$

The unconditional probability is $P(M) = 0.51$, whereas the conditional probability is $P(M \mid A) \approx 0.395$.

2.3 Section 2.3 - Multiplication Rules and Independence

Rule 5: General Multiplication Rule (GMR)

For events $A_1, A_2, \ldots, A_n$, with the conditional probabilities defined,

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Proof. Apply the Conditional Probability Rule to each conditional factor on the right:

$$\text{RHS} = P(A_1)\frac{P(A_1 \cap A_2)}{P(A_1)}\frac{P(A_1 \cap A_2 \cap A_3)}{P(A_1 \cap A_2)} \cdots \frac{P(A_1 \cap \cdots \cap A_n)}{P(A_1 \cap \cdots \cap A_{n-1})}.$$

Successive numerators and denominators cancel throughout the product:

$$\cancel{P(A_1)}\,\frac{\cancel{P(A_1 \cap A_2)}}{\cancel{P(A_1)}}\,\frac{\cancel{P(A_1 \cap A_2 \cap A_3)}}{\cancel{P(A_1 \cap A_2)}} \cdots \frac{P(A_1 \cap \cdots \cap A_n)}{\cancel{P(A_1 \cap \cdots \cap A_{n-1})}}.$$

Continuing the same cancellation through the last factor gives

$$\text{RHS} = P(A_1 \cap A_2 \cap \cdots \cap A_n) = \text{LHS}.$$

In particular, for two events,

$$\textbf{GMR, } n = 2: \qquad \text{Let } A_1 \text{ and } A_2 \text{ be two events.}$$

$$\boxed{P(A_1 \cap A_2) = P(A_1)P(A_2 \mid A_1).}$$

Example: One Male and One Female Student

A class has 40 students: 17 male and 23 female. The instructor randomly selects two students without replacement. What is the probability of selecting one male and one female?

Let S_1 and S_2 identify the first and second selections. There are two mutually exclusive orders:

$$(S_1 = M \text{ AND } S_2 = F) \quad \text{OR} \quad (S_1 = F \text{ AND } S_2 = M).$$

Hence

$$\begin{aligned} P(\text{one male and one female}) &= P\big[(\{S_1 = M\} \cap \{S_2 = F\}) \cup (\{S_1 = F\} \cap \{S_2 = M\})\big] \\ &\overset{\text{SAR}}{=} P(S_1 = M \cap S_2 = F) + P(S_1 = F \cap S_2 = M) \\ &\overset{\text{GMR}}{=} P(S_1 = M)P(S_2 = F \mid S_1 = M) \\ &\qquad + P(S_1 = F)P(S_2 = M \mid S_1 = F) \\ &= \frac{17}{40} \cdot \frac{23}{39} + \frac{23}{40} \cdot \frac{17}{39} \\ &= \boxed{\frac{2 \cdot 17 \cdot 23}{40 \cdot 39}}. \end{aligned}$$

Independent Events

Events A and B are **independent** if

$$\boxed{P(B \mid A) = P(B).}$$

Knowing that A occurred does not change the probability that B occurs.

Mutually exclusive and **independent** describe different relationships:

- Mutually exclusive events cannot occur at the same time.
- For independent events, knowing that one occurred does not change the probability of the other.

Example: Roll a Fair Die Twice

Define

$$A = \{\text{the first roll is even}\},$$
$$B = \{\text{the first roll is odd}\},$$
$$C = \{\text{the second roll is odd}\}.$$

A and B are mutually exclusive. In contrast, A and C are *not* mutually exclusive: they can occur at the same time. They are independent, since

$$P(C \mid A) = P(C) = \frac{3}{6}.$$

Rule 6: Special Multiplication Rule (SMR)

For independent events $A_1, A_2, \ldots, A_n$, the General Multiplication Rule becomes

$$\begin{aligned} P(A_1 \cap \cdots \cap A_n) &\overset{\text{GMR}}{=} P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2) \\ &\quad \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}) \\ &\overset{\text{Ind.}}{=} \boxed{P(A_1)P(A_2)\cdots P(A_n)}. \end{aligned}$$

Three Useful Patterns

Complement Rule:

$$P(A) = 1 - P(A^c).$$

Special Addition Rule, for mutually exclusive events:

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i), \qquad \text{OR} \longrightarrow \text{SUM}.$$

Special Multiplication Rule, for independent events:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i), \qquad \text{AND} \longrightarrow \text{PRODUCT}.$$

Example 2.3.2: Two Children with Brown Eyes

For the independent events that the first and second child have brown eyes, each with probability $3/4$,

$$\begin{aligned} P(C_1\colon \text{BE} \ \cap \ C_2\colon \text{BE}) &\overset{\text{SMR}}{=} P(C_1\colon \text{BE})P(C_2\colon \text{BE}) \\ &= \frac{3}{4} \cdot \frac{3}{4} = \boxed{\frac{9}{16}}. \end{aligned}$$

Here, BE denotes brown eyes.

Example 2.3.4: A Primary and Two Secondary Systems

Consider a primary system and two secondary systems. Let PSOS, 1SSOS, and 2SSOS denote the events that the primary, first secondary, and second secondary systems operate successfully.

Each system succeeds with probability 0.9, independently. The probability that **all three operate successfully** is

$$\begin{aligned} P(\text{PSOS} \cap \text{1SSOS} \cap \text{2SSOS}) &\overset{\text{SMR}}{=} P(\text{PSOS})P(\text{1SSOS})P(\text{2SSOS}) \\ &= (0.9) \cdot (0.9) \cdot (0.9). \end{aligned}$$